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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
May 2025

EE1001-2 – BASIC ELECTRICAL ENGINEERING
(For AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **Capacitor does not allow the sudden change of**
A) Current
B) Power
C) Voltage
D) Frequency
2. **Which quantity consists of a unit 1 KWh**
A) Energy
B) Time
C) Power
D) Charge
3. **KCL works on the principle of which of the following**
A) Law of conservation of energy
B) Law of conservation of charge
C) Conservation of momentum
D) Conservation of potential energy
4. **How many cycles will an AC signal make in 2 seconds if its frequency is 100 Hz?**
A) 50
B) 200
C) 100
D) 150
5. **In series RLC circuit, voltage across resistor, inductor and capacitor are 5V, 2V and 2V respectively. Find total voltage**
A) 9V
B) 2V
C) 4V
D) 5V
6. **Which of the following can vary with AC, but never with DC?**
A) Power
B) Voltage
C) Frequency
D) Amplitude
7. **In ac circuit the ratio of kW/KVA is**
A) Power factor
B) Load factor
C) Form factor
D) Peak factor
8. **What is the angle between the line voltage and the phase voltage in a star connection?**
A) 60 degrees
B) 90 degrees
C) 0 degrees
D) 30 degrees
9. **The back emf in a DC motor is**
A) Inversely proportional to the speed of the motor
B) Directly proportional to the speed of the motor
C) Independent of the speed of the motor
D) Equal to the supply voltage
10. **A DC series motor has**
A) Low starting torque
B) Constant speed
C) High starting torque
D) Low speed regulation
11. **In a transformer, the primary and secondary windings are**
A) Electrically connected
B) Magnetically coupled
C) Both electrically connected and magnetically coupled
D) Neither electrically connected nor magnetically coupled

12. The EMF equation of a transformer is
 A) $\frac{E_1}{E_2} = \frac{N_1}{N_2}$
 B) $\frac{E_1}{E_2} = \frac{N_2}{N_1}$
 C) $E_1 E_2 = N_1 N_2$
 D) $E_1 + E_2 = N_1 + N_2$
13. A single phase autotransformer has
 A) One winding
 B) Two windings
 C) Three windings
 D) Four windings
14. Slip in an induction motor is
 A) The ratio of synchronous speed to rotor speed
 B) The ratio of slip speed to synchronous speed
 C) The difference between synchronous speed and rotor speed
 D) The sum of synchronous speed and rotor speed
15. The rotor of a three-phase induction motor
 A) Rotates at synchronous speed
 B) Rotates at a speed slightly less than synchronous speed
 C) Rotates at a speed slightly more than synchronous speed
 D) Does not rotate
16. A rotating magnetic field is produced in a three-phase induction motor by
 A) Single-phase AC supply
 B) Three-phase AC supply
 C) DC supply
 D) All of these supply
17. What does MCB stand for?
 A) Miniature Circuit Breaker
 B) Main Control Board
 C) Motor Control Box
 D) Maximum Current Breaker
18. What is the advantage of PMSM over induction motors?
 A) Simpler design
 B) Larger size
 C) Lower cost
 D) Higher efficiency
19. Buck converter is also called as _____
 A) Step up converter
 B) Step Up - Step down converter
 C) Step down converter
 D) Step down - step up converter
20. What is necessary to provide a path to the leakage current to ground and protect personnel from the danger of shock?
 A) Fuse
 B) Earthing
 C) Circuit Breaker
 D) Insulator

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I

1. a) Define average value of alternating quantity. Also mention the average value for sinusoidal ac quantity for one cycle.
 b) Using mesh analysis, find the voltage drop across the 3Ω resistor in Fig. 1b

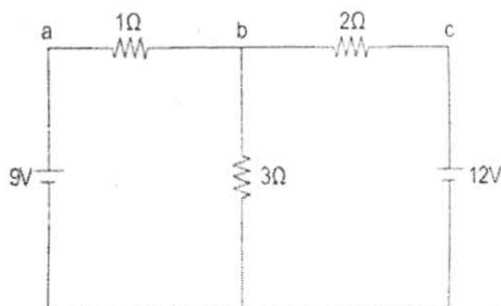


fig. 1b

- c) Deduce the expression for active power in series RL circuit. Also represent instantaneous power using wave forms.

Marks	BT*	CO*	PO*
04	L*1	1	1,2

06	L3	1	1,2
06	L2	2	1,2

2. a) Define power factor and define various forms of power in single phase circuit.
- b) Use nodal analysis to find the voltage across and current through 2Ω resistor in Figure. 2b

04 L3 1 1,2

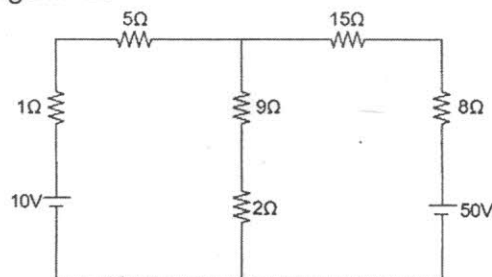


Figure 2b

- c) Deduce the relation between line and phase parameters of three phase star connection.
3. a) List the advantages of three phase connection over single phase connection.
- b) A pure resistance of 50Ω is in series with a capacitance of $50\mu\text{F}$. The series combination is connected across a 230V, 50Hz supply. Find impedance, current and power factor of the circuit.
- c) Define root mean square value of an alternating quantity. Derive an expression for RMS value of an alternating current.

06 L3 1 1,2

06 L2 2 1,2

04 L1 2 1,2

06 L3 2 1,2

06 L2 1 1,2

Unit – II

4. a) State Faradays laws of electromagnetic induction.
- b) Discuss the working principle of three phase induction motor.
- c) The primary and secondary windings of a 100 kVA transformer have resistances of 0.20Ω and 0.001Ω respectively. The primary and secondary voltages are 2000V and 200 V respectively and the core loss is 1kW. Calculate the efficiency of the transformer at full load and unity power factor.
5. a) A four pole DC Motor is connected to a 200V DC supply and takes an armature current of 50A. The resistance of the armature is 0.2Ω . The armature is lap wound with 200 conductors and useful flux per pole is 0.02 Wb. Calculate the back emf and speed of the motor.
- b) Derive the EMF equation of a single phase transformer.
- c) In an induction motor, prove that stator magnetic field has constant magnitude and it rotates at synchronous speed.
6. a) With neat diagram explain constructional features of three phase induction motor.
- b) What is an Autotransformer? List its advantages and applications.
- c) A 6 pole 3 phase induction motor operates from a supply whose frequency is 50Hz. Calculate: i) The speed at which the magnetic field of the stator is rotating. ii) The speed of the rotor when the slip is 0.02 iii) Frequency of the rotor currents when the slip is 0.04.

04 L1 3 1,2

06 L2 4 1,2

06 L3 3 1,2

04 L3 3 1,2

06 L1 4 1,2

06 L2 3 1,2

04 L1 3 1,2

06 L2 4 1,2

06 L3 3 1,2

Unit – III

7. a) List necessity of earthing electrical system.
- b) Explain the working principle of BLDC motor.
- c) With a neat diagram, elaborate the major part of an electric vehicle.
8. a) List the necessity of UPS.
- b) Discuss the principle of operation of boost converter.
- c) What is earthing? With a neat diagram, explain plate earthing.

04 L2 5 1,2

06 L1 5 1,2

06 L2 5 1,2

04 L2 5 1,2

06 L2 5 1,2

06 L2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
